POSSESION OF MOBILES IN EXAM IS UFM PRACTICE.

Name _____ **Enrollment No.**_____

Jaypee Institute of Information Technology, Noida

T1 Examination, 2025

B. Tech /Integrated M. Tech - VI Semester

Course Title: Computer Networks and Internet of Things

Course Code: 18B11CS311

Maximum Time : 1 Hr

Maximum Marks : 20

CO1	Defining the basics of networking, components and underlying technologies
CO2	Illustrate the various key protocols in OSI model and TCP/IP protocol suite and explain various application protocols.
CO3	Identification and description of various components, architectures and protocols of Internet of Things (IoT) and their real-life problems.
CO4	Choose the appropriate network layer and data link layer protocols based on the specific requirements of the communication environment.
CO5	Explain various transport protocols and its performance enhancing mechanisms.
CO6	Determine the shortest path for the network using various routing protocols and evaluate it.

Note: Attempt all the questions.

Q1. For a network, where Host A is connected to Host B via R1 (Router), S1 (Switch) and R2 (Router) sequentially. Determine how many times each packet has to visit the various layers of the TCP/IP during a transmission from Host A to Host B.

[CO1 (Remembering), 3 Marks]

Q2. You visit the website *www.pizza.com* using a web browser. The website consists of a base HTML page and multiple embedded objects. Before fetching the page, your browser needs to resolve the domain name using multiple DNS servers and iterative query. To resolve *www.pizza.com*, your browser must contact n DNS servers sequentially., each requiring 1 RTT. The base HTML page size = 40,000 bits.

Embedded objects and their sizes:

www.pizza.com/pizza_bannner.jpg = 20,000 bits
www.pizza.com/cheese.jpg = 10,000 bits
www.pizza.com/pepperoni.jpg = 15,000 bits
www.pizza.com/order_button.png = 5,000 bits
www.pizza.com/menu.txt = 5,000 bits
www.pizza.com/veggie_special.jpg = 30,000 bits

Your browser uses the HTTP protocol to download the base page and the embedded objects. Make the following assumptions:

- At most10,000 bits of data fits into a single packet. You can ignore the overhead of any headers or framing.
- You must first download the entire base page before you can start fetching the embedded images.
- HTTP requests are 5,000 bits in size.
- Any new connection to a machine requires a connection-establishment handshake.
- For this problem, you do not need to worry about closing connections, and you can ignore the delay introduced in acknowledging the final data packet sent by the server to your browser.
- All senders use windows of 20,000 bits.
- No packets are lost.

(a) For the initial transfer of the homepage (including DNS resolution with n DNS servers), how many RTTs are required, and what happens during each RTT?

(b) How quickly (in terms of RTTs) can your browser download the base page and all embedded objects if it uses:

- One connection per item, with up-to 4 concurrent connections
- A single persistent, non-pipelined connection
- A single pipelined connection

[CO2 (Understanding), 2+3 = 5 Marks)

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Q3. A 500 KB file is being transmitted from Host A to Host B through two packet switches using store-and-forward packet switching. The network operates under the following conditions:

Packet size (including header) = 10 KB

Transmission rate:

Between Host A and Switch 1 = 10 Mbps, between switch 1 and Switch 2 = 5 Mbps, between Switch 2 and Host B = 2 Mbps

Propagation delay for each link = 10 ms, Processing delay at each switch = 2ms

Average queuing delay:

At switch 1 = 4 ms per packet, At switch 2 = 6 ms per packet

a. Calculate the total transmission time for a single packet from Host A to Host B, including all delays.
b. Determine the total end-to-end delay for the entire 500 KB file to reach Host B, considering store-and-forward switching, propagation delay, queuing delay, and processing delay.
c. Find the bottleneck link in the transmission process and analyze how increasing its bandwidth to 4 Mbps would affect the overall delay.

[CO2 (Understanding), 1+1+2 = 4 Marks]

Q4. A client computer at jiit.128.edu wants to download a 10 MB file from a remote FTP server (jiit.62.noida.edu). The client first needs to resolve the hostname via the DNS iterative resolution process, establish an FTP session, and then perform the file transfer.

DNS Resolution:

RTT to the local DNS server = 10 ms

RTT to the ROOT DNS server = 20 ms

RTT to the TLD server = 25 ms

RTT to the authoritative server = 30 ms

Transmission time for DNS response = 5 ms per query

DNS operates in persistent mode (each query does not require a new connection)

FTP file transfer (passive mode):

RTT for FTP control connection setup = 30 ms

FTP data connection setup RTT = 30 ms

File size = 10 MB

FTP data transfer rate = 4 Mbps

Propagation delay is negligible

a. Compute the total time required for DNS resolution.
b. Calculate the total time required for FTP session setup (control and data connection establishment in passive mode).
c. Determine the total time required for FTP file transfer.
d. Find the overall total time from DNS resolution to the completion of file transfer.

[CO2 (Understanding), 4 Marks]

Q5. A web server is hosting multiple websites and uses TCP for communication. The server listens on port 80 for HTTP and port 443 for HTTPS traffic. Consider a client (client X) from 192.168.1.10 opens a browser and requests www.siteA.com. Another client (client Y) from IP 192.168.1.20 opens a browser and requests www.siteB.com. Both clients connect to the same IP (10.0.0.1) using port 80 for HTTP.

a. Since both clients are connecting to the same IP (10.0.0.1) and the same port (80), how does the transport layer distinguish between them?
b. If a third client (Client Z) from 192.168.1.30 also connects to the server and requests www.siteA.com, does it share the same connection as client X? why or why not?
c. Assume the server is hosting virtual websites (i.e., multiple domains on the same server). If both client X and client Z requests www.siteA.com, but from different source IPs and ports, how does the application layer (HTTP) determine which website to serve?

[CO5 (Analyzing), 1+1+2 = 4 Marks]

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